

PANYNJ maps plan to reach net-zero emissions by 2050



Currently, only about 15% of cargo that
increase that percentage as part of PAN
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Port officials hope to



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By Julie Sneider, Senior

In 2021, the Port Authority of New York New Jersey (PANYNJ) announced an ambitious goal to reduce its carbon footprint by achieving net-zero emissions by 2050. In September, the agency unveiled “Net-Zero Roadmap,” a more than 40-step plan that explains how the port authority intends to get there.

The roadmap details PANYNJ’s plans to decarbonize its operations and facilities, or its Scope 1 and Scope 2 emissions — and then how the authority aims to work with tenants and contractors to achieve the 2050 goal by reducing Scope 3 emissions.

“What we’re looking to do by 2030 is to reduce our Scope 1 and 2 emissions by 50%,” says Deputy Port Director Mike Bozza. “There’s a number of initiatives across our broad business platform to make that 50% goal by 2030. And between 2030 and 2050, we will shift our focus to those Scope 3 emissions.”

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The PANYNJ is a massive enterprise that includes five airports, bridges and tunnels, the World Trade Center, the Port of New York and New Jersey, bus terminals and PATH, the Port Authority-Trans Hudson railroad. [In rolling out the roadmap on Sept. 19](#), PANYNJ officials also announced the agency is on track to meet its interim goals of a 35% reduction in direct emissions by 2025, and a 50% reduction by 2030.

The roadmap includes converting half of the port authority's light-duty fleet to electric vehicles by 2025 and accelerating the goal of 100% conversion by 2028; decarbonizing all buildings by 2050; phasing out fossil-fuel ground-support equipment to electric at the airports; continued transition of ship-to-shore and rail-mounted gantry cranes at the seaport — that's now 98% completed; modernizing the airport shuttle bus fleet with zero-emission alternatives by 2025; and replacing 50% of its nonemergency response medium- and heavy-duty vehicles to zero-emission alternatives by 2030, and setting a plan for the remainder.



A piece of the port authority fleet to electric by 2028. PANYNJ

The agency also has plans to replace 50% of its nonemergency response EVs. And as part of the roadmap, the agency has launched at the seaport a program that provides grants of up to \$50,000 for the replacement of engines.



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Also on the map: Expanding rail access

Another essential element on the map is how the authority will use rail to achieve its net-zero emissions target by 2050. Steps include using technology to reduce emissions from PATH trains and the AirTrain service, as well as encouraging greater usage of the ExpressRail System.

At the port, the authority is trying to increase the use of intermodal rail via ExpressRail.

“The port authority has invested over \$600 million to build out this system so that every one of our container terminals has on- or near-dock rail,” Bozza says. “The first piece of how that helps with emissions is that there’s a very short trip to move cargo from the dock to the rail and onto inland destinations.”

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ExpressRail has dedicated rail facilities — and additional support track and rail yards — for each of the port's major container terminals. Annually, the system has the capacity for 1.5 million rail lifts, according to the port authority's website. The system also provides access to CSX and Norfolk Southern Railway for inland-bound cargo.

Still, only about 15% of cargo that comes into the port goes out to inland destinations by rail, according to Bozza.

“We would love — and we strive — to increase those percentages year over year,” he says.

PANYNJ officials believe they have a good case for convincing shippers to use rail to move cargo from the port.

“About 75% of the ships that come here are using the Port of New York and New Jersey as their first port of call,” Bozza notes. “If you’re shipping products to Cleveland, Chicago or to other inland destinations, we think that’s a good reason to drop them off here. As ships get larger, they will spend more time at the dock. When things get bigger, the time it takes to get cargo on and right now our rail dwell is under 24 hours, which is a lot better than before the ship leaves the berth.”

At the same time, the port, we don’t necessarily have a landlord who’s routed,” Bozza acknowledges.

“What we’re trying to do is work very closely with the Class I railroads; we spend a tremendous amount of money on the infrastructure and work with our partners to make sure the infrastructure is performing properly,” he says. “We try to make sure that people know the advantages of shipping through here — and really try to make the experience the best it can be.”

Among the port’s recent rail infrastructure investments is the \$6.6 billion Southbound Connector project, which calls for an additional track to be added to the port's ExpressRail network in Elizabeth, which serves the Maher and APM terminals. Now in the final design phase, the project will help facilitate faster intermodal rail connections. At buildout, the connection will increase the terminal’s annual capacity by 75%, from 800,000 to more than 1.4 million containers, Tyrone Harrison, PANYNJ's manager of intermodal rail development, said in an interview with [RailPrime earlier this year](#).

Construction on the Southbound Connector project is slated to begin in fourth-quarter 2024 and conclude in 2025.



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“Every train that comes out of that [ExpressRail Elizabeth] facility has to turn north,” says Bozza. “The Southbound Connector is going to create a new exit and entrance from that facility,” which will allow trains to go south, avoid certain chokepoints and save time.

Another rail investment aimed at reducing emissions was made at New York New Jersey Rail LLC, a short line marine railroad the authority acquired in 2008. NYNJR operates the rail flow between the Greenville Yard in Jersey City, New Jersey, and 65th Street Yard in Brooklyn, New York. It provides the only remaining car float service in New York Harbor, and also provides warehouse switching services at Port Jersey. The short line connects with CSX and NS via Conrail West-of-Hudson, and with New York & Atlantic Railway East-of-Hudson.

In 2015, the authority acquired three Tier-4i SE10B locomotives, two for the Greenville Yard and one for the 65th Street Yard.

Following PATH

Meanwhile, the authority is working on net-zero emissions for its fleet by switching to electric power — it already runs on fossil fuels.

"As the electric grid becomes greener, the system will decline on fossil fuels, and operational efficiency will improve with regenerative braking coupled with energy storage," the PANYNJ roadmap states.

The authority is now experimenting with regenerative braking on PATH trains to capture and reuse energy. And since rail in general already has a smaller carbon footprint, the authority will take steps to encourage more people and businesses to use it. That means finding ways to make sure riders experience pleasant and timely train trips.

In 2019, PATH served 82.2 million riders, according to the roadmap report. While ridership is still lower than those pre-pandemic levels, it is bouncing back. To encourage the upward trend, the authority set in motion a plan to reduce delays and improve the rider experience by replacing older rail cars.

In 2022, PATH began receiving the first of 72 new cars and when fully in service by late 2024, they will expand the current fleet by more than 20%. PATH also began to phase in nine-car trains to increase capacity on the



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Newark-World Trade Center line. By 2030, the authority expects to develop more ways to expand service and increase off-peak train usage.

All those rail-related elements are part of the broader roadmap the authority is sharing with various shipping industry stakeholders, including the Council of Port Performance. Made up of executives from the marine terminal operators, Class I's and the trucking community, the group oversees the port's programs and initiatives to help improve efficiency and service reliability. Bozza hopes raising awareness of the authority's Net-Zero Roadmap will encourage others in the shipping community to take actions that reduce Scope 3 emissions.

"Through our work with that group, we are working to make the entire supply chain — what's in our control and what's outside our control — as efficient as possible," Bozza says.



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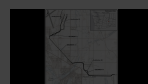
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